

Dynamic Texture Recognition Using Local Binary Patterns with an Application to Facial Expressions

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Abstract—Dynamic texture (DT) is an extension of texture to the temporal domain. Description and recognition of DTs have attracted growing attention. In this paper, a novel approach for recognizing DTs is proposed and its simplifications and extensions to facial image analysis are also considered. First, the textures are modeled with volume local binary patterns (VLBP), which are an extension of the LBP operator widely used in ordinary texture analysis, combining motion and appearance. To make the approach computationally simple and easy to extend, only the co-occurrences of the local binary patterns on three orthogonal planes (LBP-TOP) are then considered. A block-based method is also proposed to deal with specific dynamic events such as facial expressions in which local information and its spatial locations should also be taken into account. In experiments with two DT databases, DynTex and Massachusetts Institute of Technology (MIT), both the VLBP and LBP-TOP clearly outperformed the earlier approaches. The proposed block-based method was evaluated with the Cohn-Kanade facial expression database with excellent results. The advantages of our approach include local processing, robustness to monotonic gray-scale changes, and simple computation.

Index Terms—Temporal texture, motion, facial image analysis, facial expression, local binary pattern.

1 INTRODUCTION

DYNAMIC or temporal textures are textures with motion [1]. They encompass the class of video sequences that exhibit some stationary properties in time [2]. There are lots of dynamic textures (DTs) in the real world, including sea waves, smoke, foliage, fire, shower, and whirlwind. Description and recognition of DT are needed, for example, in video retrieval systems, which have attracted growing attention. Because of their unknown spatial and temporal extent, the recognition of DT is a challenging problem compared with the static case [3]. Polana and Nelson classified visual motion into activities, motion events, and DTs [4]. Recently, a brief survey of DT description and recognition was given by Chetverikov and Péteri [5].

Key issues concerning DT recognition include:

1. combining motion features with appearance features;
2. processing locally to catch the transition information in space and time, for example, the passage of burning fire changing gradually from a spark to a large fire;
3. defining features that are robust against image transformations such as rotation;
4. insensitivity to illumination variations;
5. computational simplicity; and
6. multiresolution analysis. To our knowledge, no previous method satisfies all of these requirements.

To address these issues, we propose a novel, theoretically and computationally simple approach based on local binary patterns. First, the textures are modeled with volume local binary patterns (VLBP), which are an extension of the local binary patterns (LBP) operator widely used in ordinary texture analysis [6], combining the motion and appearance. The texture features extracted in a small local neighborhood of the volume are not only insensitive with respect to translation and rotation, but also robust with respect to monotonic gray-scale changes caused, for example, by illumination variations. To make the VLBP computationally simple and easy to extend, only the co-occurrences on three separated planes are then considered. The textures are modeled with concatenated Local Binary Pattern histograms from Three Orthogonal Planes (LBP-TOP). The circular neighborhoods are generalized to elliptical sampling to fit the space-time statistics.

As our approach involves only local processing, we are allowed to take a more general view of DT recognition, extending it to specific dynamic events such as facial expressions. A block-based approach combining pixel-level, region-level, and volume-level features is proposed for dealing with such nontraditional DTs in which local information and its spatial locations should also be taken into account. This will make our approach a highly valuable tool for many potential computer vision applications. For example, the human face plays a significant role in verbal and nonverbal communication. Fully automatic and real-time facial expression recognition could find many applications, for instance, in human-computer interaction, biometrics, telecommunications, and psychological research. Most of the research on facial expression recognition has been based on static images [7], [8], [9], [10], [11], [12], [13]. Some research on using facial dynamics has also been carried out [14], [15], [16]; however, reliable segmentation of the lips and other moving facial parts in natural environments has proven to be a major problem.

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Our approach is completely different, avoiding error-prone segmentation.

2 RELATED WORK

Chetverikov and Péteri [5] placed the existing approaches of temporal texture recognition into five classes: methods based on optic flow, methods computing geometric properties in the spatiotemporal domain, methods based on local spatiotemporal filtering, methods using global spatiotemporal transforms and, finally, model-based methods that use estimated model parameters as features.

The methods based on optic flow [3], [4], [17], [18], [19], [20], [21], [22], [23], [24] are currently the most popular ones [5], because optic flow estimation is a computationally efficient and natural way to characterize the local dynamics of a temporal texture. Péteri and Chetverikov [3] proposed a method that combines normal flow features with periodicity features, in an attempt to explicitly characterize motion magnitude, directionality, and periodicity. Their features are rotation invariant, and the results are promising. However, they did not consider the multiscale properties of DT. Lu et al. proposed a new method using spatiotemporal multiresolution histograms based on velocity and acceleration fields [21]. Velocity and acceleration fields of different spatiotemporal resolution image sequences were accurately estimated by the structure tensor method. This method is also rotation invariant and provides local directionality information. Fazekas and Chetverikov compared normal flow features and regularized complete flow features in DT classification [25]. They concluded that normal flow contains information on both dynamics and shape.

Saisan et al. [26] applied a DT model [1] to the recognition of 50 different temporal textures. Despite this success, their method assumed stationary DTs that are well segmented in space and time and the accuracy drops drastically if they are not. Fujita and Nayar [27] modified the approach [26] by using impulse responses of state variables to identify model and texture. Their approach showed less sensitivity to nonstationarity. However, the problem of heavy computational load and the issues of scalability and invariance remain open. Fablet and Bouthemy introduced temporal co-occurrence [19], [20] that measures the probability of co-occurrence in the same image location of two normal velocities (normal flow magnitudes) separated by certain temporal intervals. Recently, Smith et al. dealt with video texture indexing using spatiotemporal wavelets [28]. Spatiotemporal wavelets can decompose motion into local and global, according to the desired degree of detail.

Otsuka et al. [29] assumed that DTs can be represented by moving contours whose motion trajectories can be tracked. They considered trajectory surfaces within 3D spatiotemporal volume data and extracted temporal and spatial features based on the tangent plane distribution. The spatial features include the directionality of contour arrangement and the scattering of contour placement. The temporal features characterize the uniformity of velocity components, the ash motion ratio, and the occlusion ratio. These features were used to classify four DTs. Zhong and Scarlaro [30] modified [29] and used 3D edges in the spatiotemporal domain. Their

DT features were computed for voxels taking into account the spatiotemporal gradient.

It appears that nearly all of the research on DT recognition has considered textures to be more or less “homogeneous,” that is, the spatial locations of image regions are not taken into account. The DTs are usually described with global features computed over the whole image, which greatly limits the applicability of DT recognition. Using only global features for face or facial expression recognition, for example, would not be effective since much of the discriminative information in facial images is local, such as mouth movements. In their recent work, Aggarwal et al. [31] adopted the Autoregressive and Moving Average (ARMA) framework of Doretto et al. [2] for video-based face recognition, demonstrating that temporal information contained in facial dynamics is useful for face recognition. In this approach, the use of facial appearance information is very limited. We are not aware of any DT-based approaches to facial expression recognition [7], [8], [9].

3 VOLUME LOCAL BINARY PATTERNS (VLBP)

The main difference between DT and ordinary texture is that the notion of self-similarity, central to conventional image texture, is extended to the spatiotemporal domain [5]. Therefore, combining motion and appearance to analyze DT is well justified. Varying lighting conditions greatly affect the gray-scale properties of DT. At the same time, the textures may also be arbitrarily oriented, which suggests using rotation-invariant features. It is important, therefore, to define features that are robust with respect to gray-scale changes, rotations, and translation. In this paper, we propose the use of VLBP (which could also be called 3D-LBP) to address these problems [32].

3.1 Basic VLBP

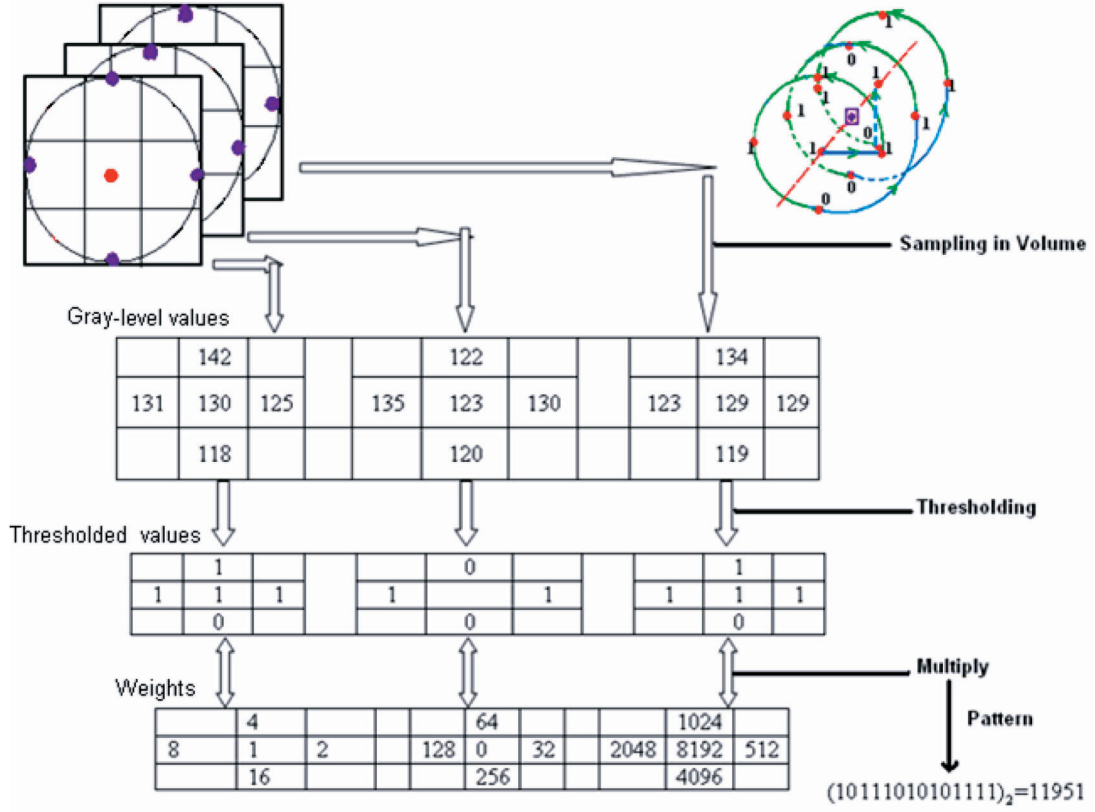
To extend LBP to DT analysis, we define DT V in a local neighborhood of a monochrome DT sequence as the joint distribution v of the gray levels of $3P + 3$ ($P > 1$) image pixels. P is the number of local neighboring points around the central pixel in one frame:

$$V = v(g_{t_c-L,c}, g_{t_c-L,0}, \dots, g_{t_c-L,P-1}, g_{t_c,c}, g_{t_c,0}, \dots, g_{t_c,P-1}, g_{t_c+L,0}, \dots, g_{t_c+L,P-1}, g_{t_c+L,c}), \quad (1)$$

where the gray value $g_{t_c,c}$ corresponds to the gray value of the center pixel of the local volume neighborhood, $g_{t_c-L,c}$ and $g_{t_c+L,c}$ correspond to the gray value of the center pixel in the previous and posterior neighboring frames with time interval L ; $g_{t,p}(t = t_c - L, t_c, t_c + L; p = 0, \dots, P - 1)$ correspond to the gray values of P equally spaced pixels on a circle of radius R ($R > 0$) in image t , which form a circularly symmetric neighbor set.

Suppose the coordinates of $g_{t_c,c}$ are (x_c, y_c, t_c) , the coordinates of $g_{t_c,p}$ are given by $(x_c + R \cos(2\pi p/P), y_c - R \sin(2\pi p/P), t_c)$ and the coordinates of $g_{t_c \pm L,p}$ are given by $(x_c + R \cos(2\pi p/P), y_c - R \sin(2\pi p/P), t_c \pm L)$. The values of the neighbors that do not fall exactly on pixels are estimated by bilinear interpolation.

To get the gray-scale invariance, the distribution is thresholded similar to that in [6]. The gray value of the volume center pixel ($g_{t_c,c}$) is subtracted from the gray values of


 Fig. 1. Procedure of $VLBP_{1,4,1}$.

the circularly symmetric neighborhood $g_{t,p}(t = t_c - L, t_c, t_c + L; p = 0, \dots, P - 1)$, giving

$$V = v(g_{t_c-L,c} - g_{t_c,c}, g_{t_c-L,0} - g_{t_c,c}, \dots, g_{t_c-L,P-1} - g_{t_c,c}, g_{t_c,c}, g_{t_c,0} - g_{t_c,c}, \dots, g_{t_c,P-1} - g_{t_c,c}, g_{t_c+L,0} - g_{t_c,c}, \dots, g_{t_c+L,P-1} - g_{t_c,c}, g_{t_c+L,c} - g_{t_c,c}). \quad (2)$$

Then, we assume that differences $g_{t,p} - g_{t,c}$ are independent of $g_{t,c}$, which allow us to factorize (2):

$$V \approx v(g_{t,c})v(g_{t_c-L,c} - g_{t_c,c}, g_{t_c-L,0} - g_{t_c,c}, \dots, g_{t_c-L,P-1} - g_{t_c,c}, g_{t_c,0} - g_{t_c,c}, \dots, g_{t_c,P-1} - g_{t_c,c}, g_{t_c+L,0} - g_{t_c,c}, \dots, g_{t_c+L,P-1} - g_{t_c,c}, g_{t_c+L,c} - g_{t_c,c}).$$

In practice, exact independence is not warranted; hence, the factorized distribution is only an approximation of the joint distribution. However, we are willing to accept a possible small loss of information as it allows us to achieve invariance with respect to shifts in gray scale. Thus, similar to LBP in ordinary texture analysis [6], the distribution $v(g_{t,c})$ describes the overall luminance of the image, which is unrelated to the local image texture and, consequently, does not provide useful information for DT analysis. Hence, much of the information in the original joint gray-level distribution (1) is conveyed by the joint difference distribution:

$$V_1 = v(g_{t_c-L,c} - g_{t_c,c}, g_{t_c-L,0} - g_{t_c,c}, \dots, g_{t_c-L,P-1} - g_{t_c,c}, g_{t_c,0} - g_{t_c,c}, \dots, g_{t_c,P-1} - g_{t_c,c}, g_{t_c+L,0} - g_{t_c,c}, \dots, g_{t_c+L,P-1} - g_{t_c,c}, g_{t_c+L,c} - g_{t_c,c}).$$

This is a highly discriminative texture operator. It records the occurrences of various patterns in the neighborhood of each pixel in a $(2(P+1) + P = 3P+2)$ -dimensional histogram.

We achieve invariance with respect to the scaling of the gray scale by considering simply the signs of the differences instead of their exact values:

$$V_2 = v(s(g_{t_c-L,c} - g_{t_c,c}), s(g_{t_c-L,0} - g_{t_c,c}), \dots, s(g_{t_c-L,P-1} - g_{t_c,c}), s(g_{t_c,0} - g_{t_c,c}), \dots, s(g_{t_c,P-1} - g_{t_c,c}), s(g_{t_c+L,0} - g_{t_c,c}), \dots, s(g_{t_c+L,P-1} - g_{t_c,c}), s(g_{t_c+L,c} - g_{t_c,c})), \quad (3)$$

where $s(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$.

To simplify the expression of V_2 , we use $V_2 = v(v_0, \dots, v_q, \dots, v_{3P+1})$, and q corresponds to the index of values in V_2 orderly. By assigning a binomial factor 2^q for each sign $s(g_{t,p} - g_{t,c})$, we transform (3) into a unique $VLBP_{L,P,R}$ number that characterizes the spatial structure of the local volume DT:

$$VLBP_{L,P,R} = \sum_{q=0}^{3P+1} v_q 2^q. \quad (4)$$

Fig. 1 shows the whole computing procedure for $VLBP_{1,4,1}$. We begin by sampling neighboring points in the volume and then thresholding every point in the neighborhood with the value of the center pixel to get a binary value. Finally, we produce the VLBP code by

multiplying the thresholded binary values with weights given to the corresponding pixel and we sum up the result.

Let us assume we are given an $X \times Y \times T$ DT ($x_c \in \{0, \dots, X-1\}$, $y_c \in \{0, \dots, Y-1\}$, $t_c \in \{0, \dots, T-1\}$). In calculating $VLBP_{L,P,R}$ distribution for this DT, the central part is only considered because a sufficiently large neighborhood cannot be used on the borders in this 3D space. The basic VLBP code is calculated for each pixel in the cropped portion of the DT, and the distribution of the codes is used as a feature vector, denoted by D :

$$D = v(VLBP_{L,P,R}(x, y, t)), x \in \{[R], \dots, X-1-[R]\}, \\ y \in \{[R], \dots, Y-1-[R]\}, t \in \{[L], \dots, T-1-[L]\}.$$

The histograms are normalized with respect to volume size variations by setting the sum of their bins to unity.

Because the DT is viewed as sets of volumes and their features are extracted on the basis of those volume textons, VLBP combines the motion and appearance to describe DTs.

3.2 Rotation Invariant VLBP

DT may also be arbitrarily oriented, so they also often rotate in the videos. The most important difference between rotation in a still texture image and DT is that the whole sequence of DT rotates around one axis or multiaxes (if the camera rotates during capturing), whereas the still texture rotates around one point. We cannot therefore deal with VLBP as a whole to get a rotation invariant code, as in [6], which assumed rotation around the center pixel in the static case. We first divide the whole VLBP code in (3) into five parts:

$$V_2 = v\left([s(g_{t_c-L,c} - g_{t_c,c})], \right. \\ [s(g_{t_c-L,0} - g_{t_c,c}), \dots, s(g_{t_c-L,P-1} - g_{t_c,c})], \\ [s(g_{t_c,0} - g_{t_c,c}), \dots, s(g_{t_c,P-1} - g_{t_c,c})], \\ [s(g_{t_c+L,0} - g_{t_c,c}), \dots, s(g_{t_c+L,P-1} - g_{t_c,c})], \\ \left. [s(g_{t_c+L,c} - g_{t_c,c})]\right).$$

Then, we mark those as V_{preC} , V_{preN} , V_{curN} , V_{posN} , and V_{posC} in order, and V_{preN} , V_{curN} , and V_{posN} represent the LBP code in the previous, current, and posterior frames, respectively, whereas V_{preC} and V_{posC} represent the binary values of the center pixels in the previous and posterior frames:

$$LBP_{t,P,R} = \sum_{p=0}^{P-1} s(g_{t,p} - g_{t_c,c})2^p, t = t_c - L, t_c, t_c + L. \quad (5)$$

Using (5), we can get $LBP_{t_c-L,P,R}$, $LBP_{t_c,P,R}$, and $LBP_{t_c+L,P,R}$.

To remove the effect of rotation, we use

$$VLBP_{L,P,R}^{ri} = \min\left\{(VLBP_{L,P,R} \text{ and } 2^{3P+1}) \right. \\ + ROL(ROR(LBP_{t_c+L,P,R}, i), 2P+1) \\ + ROL(ROR(LBP_{t_c,P,R}, i), P+1) \\ + ROL(ROR(LBP_{t_c-L,P,R}, i), 1) \\ \left. + (VLBP_{L,P,R} \text{ and } 1) | i = 0, 1, \dots, P-1\right\}, \quad (6)$$

where $ROR(x, i)$ performs a circular bitwise right shift on the P -bit number x i times [6] and $ROL(y, j)$ performs a bitwise left shift on the $3P+2$ -bit number y j times. In terms of image pixels, (6) simply corresponds to rotating the

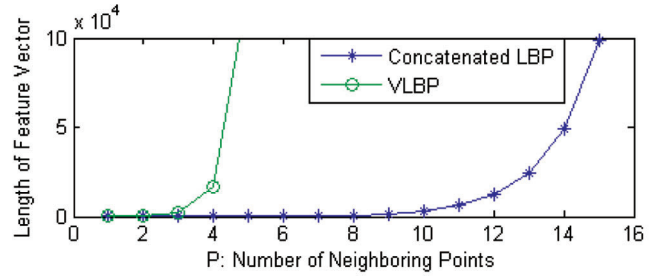


Fig. 2. The number of features versus the number of LBP codes.

neighbor set in three separate frames clockwise and this happens synchronously so that a minimal value is selected as the VLBP rotation invariant code.

For example, for the original VLBP code $(1, 1010, 1101, 1100, 1)_2$, its codes after rotating clockwise 90, 180, and 270 degrees are $(1, 0101, 1110, 0110, 1)_2$, $(1, 1010, 0111, 0011, 1)_2$, and $(1, 0101, 1011, 1001, 1)_2$, respectively. Their rotation invariant code should be $(1, 0101, 1011, 1001, 1)_2$ and not $(00111010110111)_2$ as obtained by using the VLBP as a whole.

In [6], Ojala et al. found that the vast majority of the LBP patterns in a local neighborhood are so called “uniform patterns.” A pattern is considered uniform if it contains at most two bitwise transitions from 0 to 1, or vice versa, when the bit pattern is considered circular. When using uniform patterns, all nonuniform LBP patterns are stored in a single bin in the histogram computation. This makes the length of the feature vector much shorter and allows us to define a simple version of rotation invariant LBP [6]. In the remaining sections, the superscript *riu2* will be used to denote these features, whereas the superscript *u2* means that the uniform patterns without rotation invariance are used. For example, $VLBP_{1,2,1}^{riu2}$ denotes rotation invariant $VLBP_{1,2,1}$ based on uniform patterns.

4 LOCAL BINARY PATTERNS FROM THREE ORTHOGONAL PLANES (LBP-TOP)

In the proposed VLBP, the parameter P determines the number of features. A large P produces a long histogram, whereas a small P makes the feature vector shorter but also means losing more information. When the number of neighboring points increases, the number of patterns for basic VLBP will become very large, 2^{3P+2} , as shown in Fig. 2. Due to this rapid increase, it is difficult to extend VLBP to have a large number of neighboring points and this limits its applicability. At the same time, when the time interval $L > 1$, the neighboring frames with a time variance less than L will be omitted.

To address these problems, we propose simplified descriptors by concatenating LBP on three orthogonal planes: XY, XT, and YT, considering only the co-occurrence statistics in these three directions (shown in Fig. 3). Usually, a video sequence is thought of as a stack of XY planes in axis T, but it is easy to ignore that a video sequence can also be seen as a stack of XT planes in axis Y and YT planes in axis X, respectively. The XT and YT planes provide information about the space-time transitions. With this approach, the number of bins is only $3 \cdot 2^P$, much smaller than 2^{3P+2} , as shown in Fig. 2, which makes the extension to many neighboring points easier and

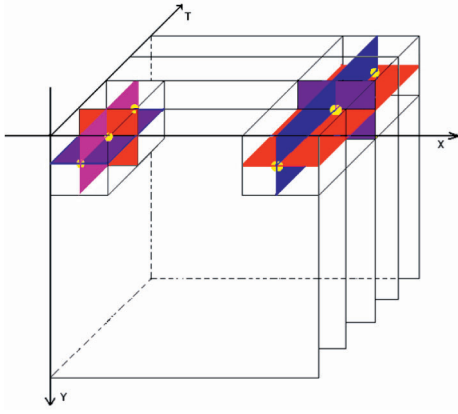


Fig. 3. Three planes in DT to extract neighboring points.

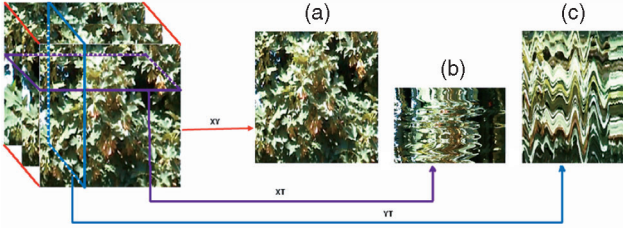


Fig. 4. (a) Image in the XY plane (400×300). (b) Image in the XT plane (400×250) in $y = 120$ (last row is pixels of $y = 120$ in first image). (c) Image in the TY plane (250×300) in $x = 120$ (first column is the pixels of $x = 120$ in first frame).

also reduces the computational complexity. There are two main differences between VLBP and LBP-TOP. First, the VLBP uses three parallel planes of which only the middle one contains the center pixel. The LBP-TOP, on the other hand, uses three orthogonal planes that intersect in the center pixel. Second, VLBP considers the co-occurrences of all neighboring points from three parallel frames, which tends to make the feature vector too long. LBP-TOP considers the feature distributions from each separate plane and then concatenates them together, making the feature vector much shorter when the number of neighboring points increases.

To simplify the VLBP for DT analysis and to keep the number of bins reasonable when the number of neighboring points increases, the proposed technique uses three instances of co-occurrence statistics obtained independently from three orthogonal planes [33], as shown in Fig. 3. Because we do not know the motion direction of textures, we also consider the neighboring points in a circle and not only in a direct line for central points in time. Compared with VLBP, not all of the volume information but only the features from three planes are applied. Fig. 4 demonstrates example images from three planes. Fig. 4a shows the image in the XY plane, Fig. 4b shows the image in the XT plane, which gave the visual impression of one row changing in time, whereas Fig. 4c describes the motion of one column in temporal space. The LBP code is extracted from the XY, XT, and YT planes, which are denoted as $XY-LBP$, $XT-LBP$, and $YT-LBP$ for all pixels and the statistics of the three different planes are obtained and then concatenated into a single histogram. The procedure is demonstrated in Fig. 5. In such a representation, DT is encoded by the $XY-LBP$, $XT-LBP$, and $YT-LBP$, whereas the appearance and motion in three directions of DT are considered, incorporating spatial domain information

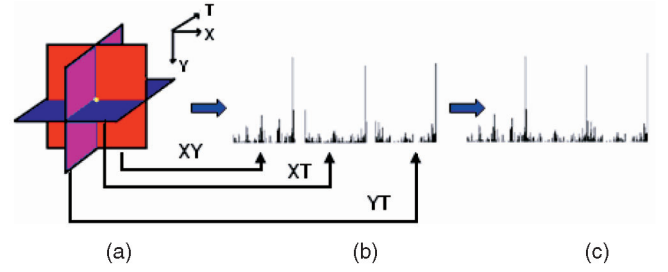


Fig. 5. (a) Three planes in DT. (b) LBP histogram from each plane. (c) Concatenated feature histogram.

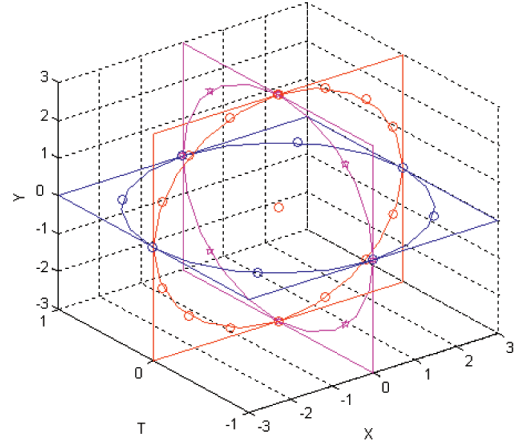


Fig. 6. Different radii and number of neighboring points on three planes.

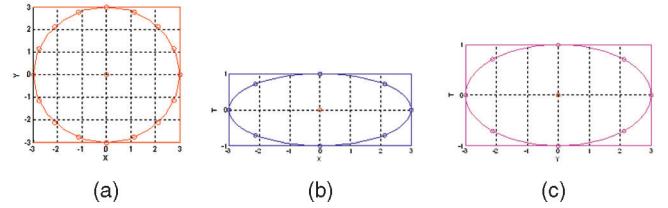


Fig. 7. Detailed sampling for Fig. 6, with $R_X = R_Y = 3$, $R_T = 1$, $P_{XY} = 16$, and $P_{XT} = P_{YT} = 8$. (a) XY plane. (b) XT plane. (c) YT plane.

($XY-LBP$) and two spatial temporal co-occurrence statistics ($XT-LBP$ and $YT-LBP$).

Setting the radius in the time axis to be equal to the radius in the space axis is not reasonable for DT. For instance, for a DT with an image resolution of more than 300×300 and a frame rate of less than 12 in a neighboring area with a radius of eight pixels in the X-axis and Y-axis, the texture might still keep its appearance; however, within the same temporal intervals in the T-axis, the texture changes drastically, especially in those DTs with high image resolution and a low frame rate. Therefore, we have different radius parameters in space and time to set. In the XT and YT planes, different radii can be assigned to sample neighboring points in space and time. With this approach, the traditional circular sampling is extended to elliptical sampling.

More generally, the radii in axes X, Y, and T and the number of neighboring points in the XY, XT, and YT planes can also be different, which can be marked as R_X , R_Y , and R_T , P_{XY} , P_{XT} , and P_{YT} , as shown in Figs. 6 and 7. The corresponding feature is denoted as $LBP-TOP_{P_{XY}, P_{XT}, P_{YT}, R_X, R_Y, R_T}$. Suppose the coordinates of the center pixel $g_{t,c}$ are (x_c, y_c, t_c) , the coordinates of

$g_{XY,p}$ are given by $(x_c - R_X \sin(2\pi p/P_{XY}), y_c + R_Y \cos(2\pi p/P_{XY}), t_c)$, the coordinates of $g_{XT,p}$ are given by $(x_c - R_X \sin(2\pi p/P_{XT}), y_c, t_c - R_T \cos(2\pi p/P_{XT}))$, and the coordinates of $g_{YT,p}$ $(x_c, y_c - R_Y \cos(2\pi p/P_{YT}), t_c - R_T \sin(2\pi p/P_{YT}))$. This is different from the ordinary LBP widely used in many papers and it extends the definition of LBP.

Let us assume we are given an $X \times Y \times T$ DT ($x_c \in \{0, \dots, X-1\}$, $y_c \in \{0, \dots, Y-1\}$, $t_c \in \{0, \dots, T-1\}$). In calculating $LBP - TOP_{P_{XY}, P_{XT}, P_{YT}, R_X, R_Y, R_T}$ distribution for this DT, the central part is only considered because a sufficiently large neighborhood cannot be used on the borders in this 3D space.

A histogram of the DT can be defined as

$$H_{i,j} = \sum_{x,y,t} I\{f_j(x,y,t) = i\}, \quad (7)$$

$$i = 0, \dots, n_j - 1; j = 0, 1, 2$$

in which n_j is the number of different labels produced by the LBP operator in the j th plane ($j = 0 : XY, 1 : XT$ and $2 : YT$), $f_i(x,y,t)$ expresses the LBP code of central pixel (x,y,t) in the j th plane and $I\{A\} = \begin{cases} 1, & \text{if } A \text{ is true;} \\ 0, & \text{if } A \text{ is false.} \end{cases}$

When the DTs to be compared are of different spatial and temporal sizes, the histograms must be normalized to get a coherent description:

$$N_{i,j} = \frac{H_{i,j}}{\sum_{k=0}^{n_j-1} H_{k,j}}. \quad (8)$$

In this histogram, a description of DT is effectively obtained based on LBP from three different planes. The labels from the XY plane contain information about the appearance, and, in the labels from the XT and YT planes, co-occurrence statistics of motion in horizontal and vertical directions are included. These three histograms are concatenated to build a global description of DT with the spatial and temporal features.

5 LOCAL DESCRIPTORS FOR FACIAL IMAGE ANALYSIS

Local texture descriptors have gained increasing attention in facial image analysis due to their robustness to challenges such as pose and illumination changes. Recently, Ahonen et al. proposed a novel facial representation for face recognition from static images based on LBP features [34], [35]. In this approach, the face image is divided into several regions (blocks) from which the LBP features are extracted and concatenated into an enhanced feature vector. This approach is proving to be a growing success. It has been adopted and further developed by many research groups and has been successfully used for face recognition, face detection, and facial expression recognition [35]. All of these have applied LBP-based descriptors only for static images, that is, they do not utilize temporal information as proposed in this paper.

In this section, a block-based approach for combining pixel-level, region-level, and temporal information is proposed. Facial expression recognition is used as a case study, but a similar approach could be used for recognizing other specific dynamic events such as faces from video, for example. The goal of facial expression recognition is to determine the emotional state of the face, for example,

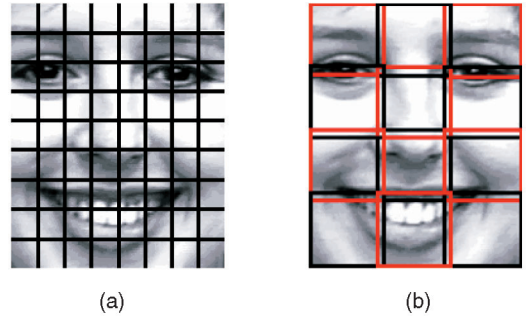


Fig. 8. (a) Nonoverlapping blocks (9×8). (b) Overlapping blocks (4×3 , overlap size = 10).

happiness, sadness, surprise, neutral, anger, fear, and disgust, regardless of the identity of the face.

Most of the proposed methods use a mug shot of each expression that captures the characteristic image at the apex [10], [11], [12], [13]. However, according to psychologists [36], analyzing a sequence of images produces more accurate and robust recognition of facial expressions. Psychological studies have suggested that facial motion is fundamental to the recognition of facial expressions. Experiments conducted by Bassili [36] demonstrate that humans are better at recognizing expressions from dynamic images as opposed to mug shots. For using dynamic information to analyze facial expression, several systems attempt to recognize fine-grained changes in the facial expression. These are based on the Facial Action Coding System (FACS) developed by Ekman and Friesen [37] for describing facial expressions by action units (AUs). Some papers attempt to recognize a small set of prototypical emotional expressions, that is, joy, surprise, anger, sadness, fear, and disgust, for example [14], [15], [16]. Yeasin et al. [14] used the horizontal and vertical components of the flow as features. At the frame level, the k -nearest neighbor (NN) rule was used to derive a characteristic temporal signature for every video sequence. At the sequence level, discrete Hidden Markov Models (HMMs) were trained to recognize the temporal signatures associated with each of the basic expressions. This approach cannot deal with illumination variations, however. Aleksic and Katsaggelos [15] proposed facial animation parameters as features describing facial expressions and utilized multistream HMMs for recognition. The system is complex, making it difficult to perform in real time. Cohen et al. [16] introduced a Tree-Augmented-Naive Bayes classifier for recognition. However, they only experimented on a set of five people, and the accuracy was only around 65 percent for person-independent evaluation.

Considering the motion of the facial region, we propose region-concatenated descriptors on the basis of the algorithm in Section 4 for facial expression recognition. An LBP description computed over the whole facial expression sequence encodes only the occurrences of the micropatterns without any indication about their locations. To overcome this effect, a representation in which the face image is divided into several nonoverlapping or overlapping blocks is introduced. Fig. 8a depicts nonoverlapping 9×8 blocks and Fig. 8b depicts overlapping 4×3 blocks with an overlap of 10 pixels, respectively. The LBP-TOP histograms in each block are computed and concatenated into a single histogram, as Fig. 9 shows. All features extracted from each block volume

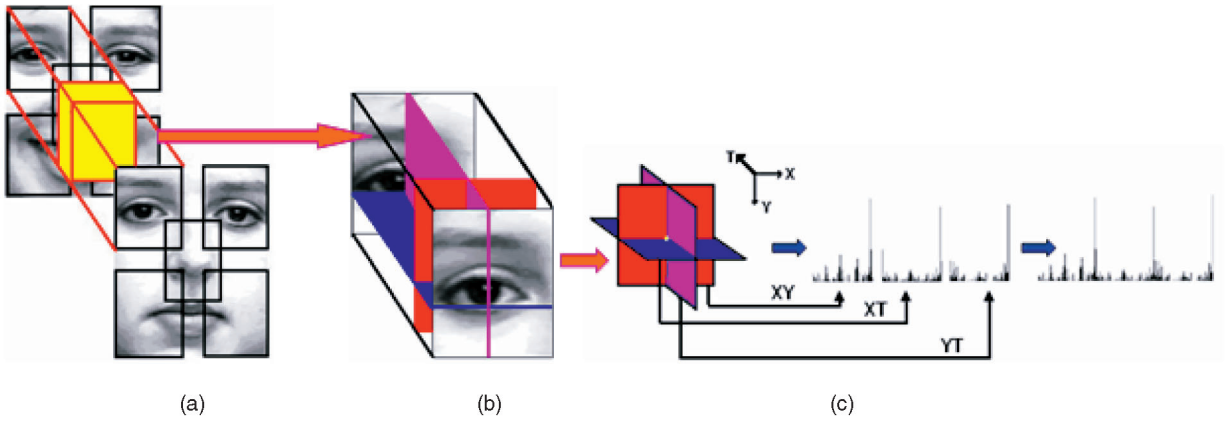


Fig. 9. Features in each block volume. (a) Block volumes. (b) LBP features from three orthogonal planes. (c) Concatenated features for one block volume with appearance and motion.

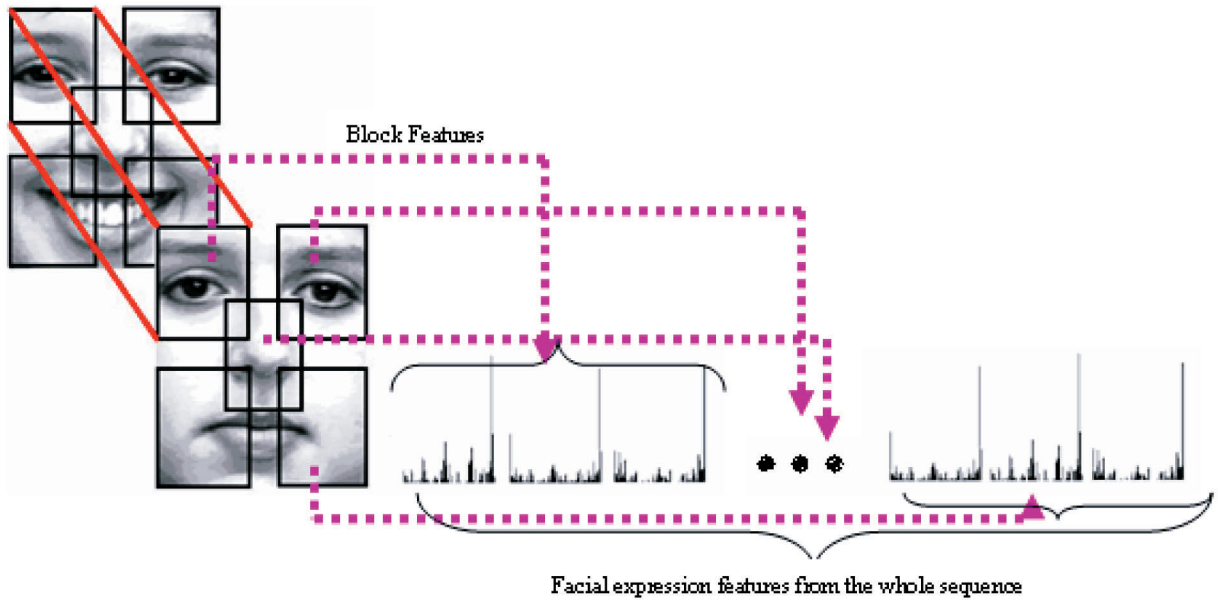


Fig. 10. Facial expression representation.

are connected to represent the appearance and motion of the facial expression sequence, as shown in Fig. 10.

In this way, we effectively have a description of the facial expression on three different levels of locality. The labels (bins) in the histogram contain information from three orthogonal planes, describing appearance and temporal information at the pixel level. The labels are summed over a small block to produce information on a regional level expressing the characteristics for the appearance and motion in specific locations and all information from the regional level is concatenated to build a global description of the face and expression motion.

Ojala et al. noticed that, in their experiments with texture images, uniform patterns account for slightly less than 90 percent of all patterns when using the (8, 1) neighborhood and around 70 percent in the (16, 2) neighborhood [6]. In our experiments, for histograms from the two temporal planes, the uniform patterns account for slightly less than those from the spatial plane but also follow the rule that most of the patterns are uniform. Therefore, the following is the notation for the LBP operator: $LBP - TOP_{P_{XY}, P_{XT}, P_{YT}, R_X, R_Y, R_T}^{u2}$ is used. The subscript represents using the operator in a

$(P_{XY}, P_{XT}, P_{YT}, R_X, R_Y, R_T)$ neighborhood. Superscript $u2$ stands for using only uniform patterns and labeling all remaining patterns with a single label.

6 EXPERIMENTS

The new large DT database DynTex was used to evaluate the performance of our DT recognition methods. Additional experiments with the widely used Massachusetts Institute of Technology (MIT) data set [1], [3] were also carried out. In facial expression recognition, the proposed algorithms were evaluated on the Cohn-Kanade Facial Expression Database [9].

6.1 DT Recognition

6.1.1 Measures

After obtaining the local features on the basis of different parameters of L , P , and R for VLBP or P_{XY} , P_{XT} , P_{YT} , R_X , R_Y , and R_T for LBP-TOP, a leave-one-group-out classification test was carried out for DT recognition based on the nearest class. If one DT includes m samples, we separate all DT samples into m groups, evaluate performance by letting



Fig. 11. DynTex database.

each sample group be unknown, and train on the remaining $m - 1$ sample groups. The mean VLBP features or LBP-TOP features of all the $m - 1$ samples are computed as the feature for the class. The omitted sample is classified or verified according to its difference with respect to the class using the k -NN method ($k = 1$).

In classification, the dissimilarity between a sample and model feature distribution is measured using the log-likelihood statistic: $L(S, M) = -\sum_{b=1}^B S_b \log M_b$, where B is the number of bins and S_b and M_b correspond to the sample and model probabilities at bin b , respectively. Other dissimilarity measures like histogram intersection or Chi square distance could also be used.

When the DT is described in the XY, XT, and YT planes, it can be expected that some of the planes contain more useful information than others in terms of distinguishing between DTs. To take advantage of this, a weight can be set for each plane based on the importance of the information it contains. The weighted log-likelihood statistic is defined as $L_w(S, M) = -\sum_{i,j} (w_j S_{j,i} \log M_{j,i})$ in which w_j is the weight of plane j .

6.1.2 Multiresolution Analysis

By altering L , P , and R for VLBP, P_{XY} , P_{XT} , P_{YT} , R_X , R_Y , and R_T for LBP-TOP, we can realize operators for any quantization of the time interval, the angular space, and spatial resolution. Multiresolution analysis can be accomplished by combining the information provided by multiple operators of varying (L, P, R) and $(P_{XY}, P_{XT}, P_{YT}, R_X, R_Y, R_T)$.

The most accurate information would be obtained by using the joint distribution of these codes [6]. However, such a distribution would be overwhelmingly sparse with any reasonable size of image and sequence. For example, the joint distribution of $VLBP_{1,4,1}^{riu2}$, $VLBP_{2,4,1}^{riu2}$, and $VLBP_{2,8,1}^{riu2}$ would contain $16 \times 16 \times 28 = 7,168$ bins. Therefore, only the marginal distributions of the different operators are considered, even though the statistical independence of the outputs of the different VLBP operators or simplified concatenated bins from three planes at a central pixel cannot be warranted.

In our study, we perform straightforward multiresolution analysis by defining the aggregate dissimilarity as the sum of the individual dissimilarity between the different operators on the basis of the additivity property of the log-likelihood statistic [6]: $L_N = \sum_{n=1}^N L(S^n, M^n)$, where N is the number of operators, and S^n and M^n correspond to the sample and model histograms extracted with operator n ($n = 1, 2, \dots, N$).

6.1.3 Experimental Setup

The DynTex data set (<http://www.cwi.nl/projects/dyntex/>) is a large and varied database of DTs.

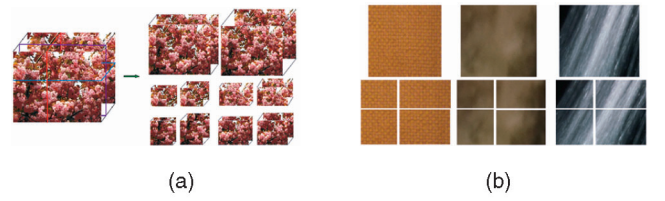


Fig. 12. (a) Segmentation of DT sequence. (b) Examples of segmentation in space.

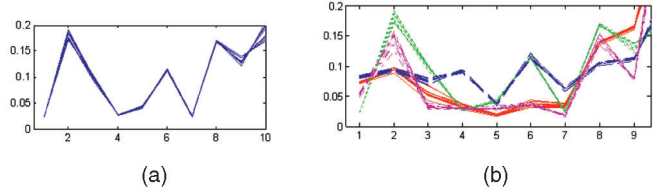
Fig. 13. Histograms of DTs. (a) Histograms of up-down tide with 10 samples for $VLBP_{2,2,1}^{riu2}$. (b) Histograms of four classes each with 10 samples for $VLBP_{2,2,1}^{riu2}$.

Fig. 11 shows example DTs from this data set. The image size is 400×300 .

In the experiments on the DynTex database, each sequence was divided into eight nonoverlapping subsets but not half in X , Y , and T . The segmentation position in volume was selected randomly. For example, in Fig. 12, we select the transverse plane with $x = 170$, the lengthways plane with $y = 130$, and the time direction with $t = 100$. These eight samples do not overlap each other and they have different spatial and temporal information. Sequences with the original size but only cut in the time direction are also included in the experiments. Therefore, we can get 10 samples of each class and all samples are different in image size and sequence length from each other. Fig. 12a demonstrates the segmentation and Fig. 12b shows some segmentation examples in space. We can see that this sampling method increases the challenge of recognition in a large database.

6.1.4 Results of VLBP

Fig. 13a shows the histograms of 10 samples of a DT using $VLBP_{2,2,1}^{riu2}$. We can see that for different samples of the same class, their VLBP codes are very similar to each other, even if they are different in spatial and temporal variation. Fig. 13b depicts histograms of four classes each with 10 samples, as in Fig. 12a. We can clearly see that the VLBP features have good similarity within classes and good discrimination between classes.

Table 1 presents the overall classification rates. The selection of optimal parameters is always a problem. Most approaches get locally optimal parameters by experiments or experience. According to our earlier studies on LBP such as [6], [10], [34], [35], the best radii are usually not bigger than three, and the number of neighboring points (P) is 2^n ($n = 1, 2, 3, \dots$). In our proposed VLBP, when the number of neighboring points increases, the number of patterns for basic VLBP will become very large: 2^{3P+2} . Due to this rapid increase, the feature vector will soon become too long to handle. Therefore, only the results for $P = 2$ and $P = 4$ are given in Table 1. Using all 16,384 bins of the basic $VLBP_{2,4,1}$ provides a 94.00 percent rate, whereas $VLBP_{2,4,1}$ with $u2$ gives a good result of 93.71 percent using only 185 bins.

TABLE 1
Results (Percent) in DynTex Data Set

	$VLBP_{1,2,1}$	$VLBP_{2,2,1}$	$VLBP_{1,4,1}$	$VLBP_{2,4,1}$
Basic	91.71 (256)	91.43 (256)	94.00 (16384)	94.00 (16384)
$u2$	87.71 (59)	90.00 (59)	93.43 (185)	93.71 (185)
ri	89.43 (144)	90.57 (144)	93.14 (4176)	95.71 (4176)
$riu2$	83.43 (10)	83.43 (10)	88.57 (16)	85.14 (16)
multi-resolution	$VLBP_{2,2,1}^{riu2} : 83.43, VLBP_{2,4,1}^{riu2} : 85.14$		$VLBP_{1,2,1}^{riu2} : 83.43, VLBP_{2,2,1}^{riu2} : 85.14$	

(Superscript $riu2$ means rotation invariant uniform patterns, $u2$ is the uniform patterns without rotation invariance, and ri represents the rotation invariant patterns. The numbers inside the parentheses demonstrate the length of corresponding feature vectors.)

When using rotation invariant $VLBP_{2,4,1}^{ri}$ (4,176 bins), we get a result of 95.71 percent. With more complex features or multiresolution analysis, better results could be expected. For example, the multiresolution features $VLBP_{2,2,1}^{riu2}$ obtain a good rate of 90 percent, better than the results from the respective features $VLBP_{2,2,1}^{riu2}$ (83.43 percent) and $VLBP_{2,4,1}^{riu2}$ (85.14 percent). However, when using multi-resolution analysis for basic patterns, the results are not improved, partly because the feature vectors are too long.

It can be seen that the basic VLBP performs very well, but it does not allow the use of many neighboring points P . A higher value for P is shown to provide better results. With uniform patterns, the feature vector length can be reduced without much loss in performance. Rotation invariant features perform almost as well as the basic VLBP for these textures. They further reduce the feature vector length and can handle the recognition of DTs after rotation.

6.1.5 Results for LBP-TOP

Table 2 presents the overall classification rates for LBP-TOP. The first three columns give the results using only one histogram from the corresponding plane; they are much lower than those from direct concatenation (fourth column) and weighed measures (fifth column). Moreover, the weighted measures of three histograms achieved better results than direct concatenation because they consider the different contributions of the features. When using only the $LBP-TOP_{8,8,1,1,1}$, we get good results of 97.14 percent with the weight [5, 2, 1] for the three histograms. Because the frame rate or the resolution in the temporal axis in the DynTex is high enough for using the same radius, it can be seen in Table 2 that results from a smaller radius in the time axis T (the last row) are as good as the same radius to the other two planes.

For selecting the weights, a heuristic approach was used, which takes into account the different contributions of the

features from the three planes. First, by computing the recognition rates for each plane separately, we get three rates $X = [x_1, x_2, x_3]$; then, we can assume that the lower the minimum rate, the smaller the relative advantage will be. For example, the recognition rate improvement from 70 percent to 80 percent is better than from 50 percent to 60 percent, even though the differences are both 10 percent. The relative advantage of the two highest rates to the lowest one can now be computed as $Y = (X - \min(X)) / ((100 - \min(X)) / 10)$. Finally, considering that the weight of the lowest rate is 1, the weights of the other two histograms can be obtained according to a linear relationship of their differences to that with the lowest rate. The following presents the final computing step and W is the generated weight vector corresponding to the three histograms:

$$Y1 = \text{round}(Y); Y2 = (Y \times ((\max(Y1) - 1)) / \max(Y) + 1); \\ W = \text{round}(Y2).$$

As demonstrated in Table 2, the larger the number of neighboring points, the better the recognition rate. For $LBP-TOP_{8,8,1,1,1}$, the result 97.14 percent is much better than that for $LBP-TOP_{4,4,1,1,1}$ (94.29 percent) and $LBP-TOP_{2,2,1,1,1}$ (90.86 percent). It should be noted, however, that $P = 4$ can be preferred in some cases because it generates a feature vector with much lower dimensionality than $P = 8$ (48 versus 768), whereas it does not suffer too much loss in accuracy (94.29 percent versus 97.14 percent). Therefore, the choice of optimal P really depends on the accuracy requirement, as well as the size of data set.

As Table 1 shows, an accuracy of 91.71 percent is obtained for VLBP using $P = 2$ with a feature vector length of 256. The length of the LBP-TOP feature vector is only 12, achieving 90.86 percent accuracy. For $P = 4$, the long VLBP feature vector (16,384) provides an accuracy of 94.00 percent, whereas the LBP-TOP achieves 94.29 percent with a feature vector of 48 elements. This clearly demonstrates that the LBP-TOP is a more effective representation for DT.

There were two kinds of commonly occurring misclassifications. One of these is the different sequences of similar classes. For example, the two steam sequences (Fig. 14a) and two parts (top left and top right in the blocks) of the steam context in Fig. 14b were discriminated into the same class. Actually, they should be thought of as the same DT. Therefore, our algorithm considering them as one class seems to prove its effectiveness in another aspect. The other one is a mixed DT, as shown in Fig. 14c, which includes more than one DT: water and shaking grass. Therefore, it shows both characteristics of these two DTs. If we think of Fig. 14a and the top-left and top-right parts of Fig. 14b as

TABLE 2
Results (Percent) in DynTex Data Set (Values in Square Brackets Are Weights Assigned for Three Sets of LBP Bins)

$LBP-TOP$	XY	XT	YT	Con	Weighted
8, 8, 8, 1, 1, 1 $u2$	92.86	88.86	89.43	94.57	96.29[4,1,1]
2, 2, 2, 1, 1, 1 $Basic$	70.86	60.86	78.00	88.86	90.86[3,1,4]
4, 4, 4, 1, 1, 1 $Basic$	94.00	86.29	91.71	93.71	94.29[6,1,5]
8, 8, 8, 1, 1, 1 $Basic$	95.14	90.86	90.00	95.43	97.14[5,2,1]
8, 8, 8, 3, 3, 3 $Basic$	90.00	91.17	94.86	95.71	96.57[1,2,5]
8, 8, 8, 3, 3, 3 $Basic$	89.71	91.14	92.57	94.57	95.14[1,2,3]

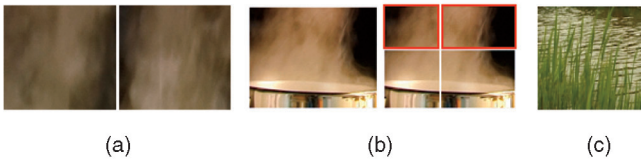


Fig. 14. Examples of misclassified sequences. (a) Steam. (b) Steam_context. (c) Water grass.

the same class, our resulting recognition rate using concatenated $LBP - TOP_{8,8,1,1,1}$ is 99.43 percent.

Our results are very good compared to the state of the art. In [25], a classification rate of 98.1 percent was reported for 26 classes of the DynTex database. However, their test and training samples were only different in the length of the sequence, but the spatial variation was not considered. This means that their experimental setup was much simpler. When we experimented using all 35 classes with samples having the original image size but only different in sequence length, a 100 percent classification rate using $VLBP_{1,8,1}^{u2}$ or using $LBP - TOP_{8,8,1,1,1}$ was obtained.

We also performed experiments on the MIT data set [1] using a similar experimental setup as with DynTex, obtaining a 100 percent accuracy both for the VLBP and LBP-TOP. None of the earlier methods has reached the same performance (see, for example, [1], [3], [25], [29]). Except for that in [3], which used the same segmentation as ours but with only 10 classes, all other papers used simpler data sets that did not include the variation in space and time.

Comparing VLBP and LBP-TOP, they both combine motion and appearance together and are robust to translation and illumination variations. Their differences are 1) the VLBP considers the co-occurrence of all the neighboring points in three frames of volume at the same time, whereas LBP-TOP only considers the three orthogonal planes, making it easy to be extended to use more neighboring information, 2) when the time interval $L > 1$, the neighboring frames with a time variance of less than L will be missed out in VLBP, but the latter method still keeps the local information from all neighboring frames, and 3) computation of the latter is much simpler with the complexity $O(XYT \cdot 2^P)$ compared to the VLBP with $O(XYT \cdot 2^{3P})$.

6.2 Facial Expression Recognition Experiments

6.2.1 Data Set

The Cohn-Kanade facial expression database [9] consists of 100 university students ranging in age from 18 to 30 years. Sixty-five percent were female, 15 percent were African-American, and 3 percent were Asian or Latino. The subjects were instructed by an experimenter to perform a series of 23 facial displays that included a single AU and combinations of AUs, six of which were based on descriptions of prototypical emotions of anger, disgust, fear, joy, sadness, and surprise. The image sequences from neutral to the target display were digitized into 640×480 or 490 pixel arrays with an 8-bit precision for gray-scale values. The video rate is 30 frames per second. For our study, 374 sequences from the data set are selected from the database for basic emotional expression recognition. The selection criterion was that any sequence to be labeled was one of the six basic emotions. The sequences came from 97 subjects, with one to six emotions per subject. The positions of the two eyes in the first frame of each



Fig. 15. Examples of in-plane rotated faces.



Fig. 16. Variation of illumination.

sequence were given manually and then these positions were used to determine the facial area for the whole sequence. The whole sequence was used to extract the proposed spatiotemporal LBP features.

Just the positions of the eyes from the first frame of each sequence were used for alignment, and this alignment was used not only for the first frame but also for the whole sequence. Though there may be some translation—in-plane rotation (as shown in Fig. 15) and out-of-plane rotation (subjects S045 and S051 in the database, permission not granted to be shown in publications) of the head—no further alignment of facial features such as alignment of the mouth was performed in our algorithm. We do not need to

1. track faces or eyes in the following frames as in [38],
2. segment eyes and outer lips [15],
3. select constant illumination [15] or perform illumination correction [14], and
4. align facial features with respect to the canonical template [14] or normalize the faces to a fixed size as done by most of the papers [11], [12], [13], [38].

In this way, our experimental setup is more realistic. Moreover, there are also some illumination and skin color variations (as shown in Fig. 16). Due to our methods' robustness to monotonic gray-level changes, there was no need for image correction by histogram equalization or gradient correction, for example, which is normally needed as a preprocessing step in facial image analysis.

6.2.2 Evaluation and Comparison

As expected, the features from the XY plane contribute less than the features extracted from the XT and YT planes. This is because the facial expression is different from ordinary DTs, its appearance features are not as important as those of the DTs. The appearance of DT can help greatly in recognition, but the appearance of a face includes both identity and expression information, which could make accurate expression classification more difficult. Features from the XT and YT planes explain more about the motion of facial muscles.

After experimenting with different block sizes and overlapping ratios, as shown in Fig. 17, we chose to use 9×8 blocks in our experiments. The best results were obtained with an overlap ratio of 70 percent of the original nonoverlapping block size. We use the overlapping ratio with the original block size to adjust and get the new overlapping ratio. Suppose the overlap ratio to the original block width is ra , after adjusting, the new ratio is ra' , and the following equation represents the ra' in height:

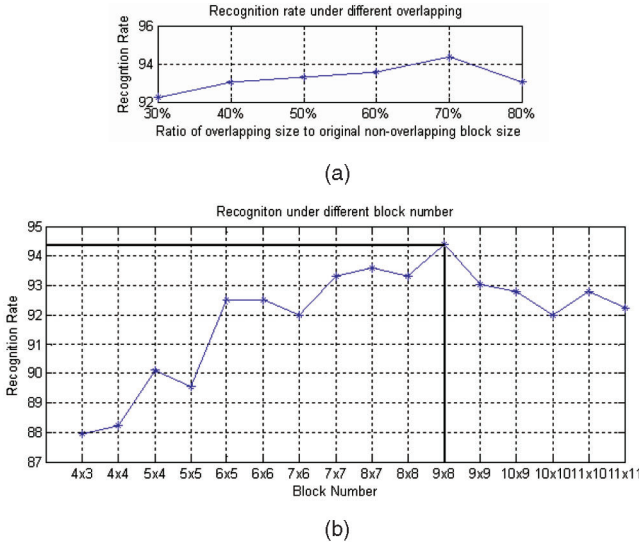


Fig. 17. Recognition with $LBP - TOP_{8,8,3,3,3}^{u2}$ under (a) different overlapping sizes and (b) number of blocks.

$$ra' = \frac{ra \cdot h/r}{\frac{(ra \cdot h \cdot (r-1))/r + h}{r}} = \frac{ra \cdot h}{(ra \cdot h \cdot (r-1))/r + h}$$

$$= \frac{ra \cdot h \cdot r}{ra \cdot h \cdot (r-1) + h \cdot r} = \frac{ra \cdot r}{ra \cdot (r-1) + r},$$

where h is the height of the block, and r is the row number of blocks. Therefore, the final overlapping ratio corresponding to 70 percent of the original block size is about 43 percent, as Table 3 shows.

For evaluation, we separated the subjects randomly into n groups of roughly equal size and did a “leave-one-group-out” cross validation, which also could be called an “ n -fold cross-validation” test scheme. They are subject independent. The same subjects did not always appear in both training and testing. The testing was therefore done with “novel faces” and was person independent.

A support vector machine (SVM) classifier was selected since it is well founded in statistical learning theory and has been successfully applied to various object detection tasks in computer vision. Since SVM is only used for separating two sets of points, the six-expression classification problem is decomposed into 15 two-class problems (happiness-surprise, anger-fear, sadness-disgust, and so forth), then, a voting scheme is used to accomplish recognition. Sometimes, more than one class gets the highest number of votes. In this case, 1-NN template matching is applied to these classes to reach the final result. This means that, in training, the spatiotemporal LBP histograms of face sequences belonging to a given class are averaged to generate a histogram template for that class. In recognition, a nearest-neighbor classifier is adopted, that is, the VLBP or LBP-TOP histogram of the input sequence sample s is classified to the nearest class template n : $L(s, n) < L(s, c)$ for all $c \neq n$ (c and n are the indices of the expression classes having the same highest number of votes in the SVM classifier).

Here, after the comparison of linear, polynomial, and RBF kernels in the experiments, we use the second degree polynomial kernel function $K(\cdot, \cdot)$, defined by $K(x, t_i) = (1 + x \cdot t_i)^2$, which provided the best results.

We did some experiments using $P = 2, 4, 8, 16$ with different radii, overlapping rates, and number of blocks. The value $P = 8$ gave better results than the other values.

TABLE 3
Overlapping Ratio before and after Overlapping Adjust for 9×8 Blocks

$ra(\%)$		30	40	50	60	70	80
$ra'(\%)$	height	23.7	29.5	34.6	39.1	43.2	46.8
	width	23.8	29.6	34.8	39.3	43.4	47.1

Then, lots of the facial expression recognition experiments were performed with the value $P = 8$, radii 1 and 3, overlapping rates from 30-80 percent, and the number of blocks with $m \times n$ ($m = 4, \dots, 11; n = m - 1, m$). The experimental results given in Fig. 18 and Table 4 were selected to give a concise presentation.

Table 4 presents the results of a two-fold cross validation for each expression using different features. As we can see, a larger number of features do not necessarily provide better results; for example, the recognition rate of $LBP - TOP_{16,16,16,3,3,3}^{u2}$ is lower than that of the $LBP - TOP_{8,8,3,3,3}^{u2}$. With a smaller number of features, $LBP - TOP_{8,8,3,3,3}^{u2}$ (177 patterns per block) outperforms $VLBP_{3,2,3}$ (256 patterns per block). The combination of $LBP - TOP_{8,8,3,3,3}^{u2}$ and $VLBP_{3,2,3}$ (last row) does not improve much compared to the results with separate features. It also can be seen that expressions labeled as surprise, happiness, sadness, and disgust are recognized with very high accuracy (94.5-100 percent), whereas fear and anger present a slightly lower recognition rate.

Fig. 18 shows a comparison to some other dynamic analysis approaches using the recognition rates given in each paper. It should be noted that the results are not directly comparable due to different experimental setups, preprocessing methods, the number of sequences used, and so forth, but they still give an indication of the discriminative power of each approach. In the left part of the figure are the detailed results for every expression. Our method outperforms the other methods in almost all cases, being the clearest winner in the case of fear and disgust. The right part of the figure gives the overall evaluation. As we can see, our algorithm works best with the greatest number of people and sequences. Table 5 compares our method with other static and dynamic analysis methods in terms of the number of people, the number of sequences and expression classes, with different measures, providing the overall results obtained with the Cohn-Kanade facial expression database. We can see that with the experiments on the greatest number of people and sequences, our results are the best ones compared not only to the dynamic methods but also to the static ones.

Table 6 summarizes the confusion matrix obtained using a 10-fold cross-validation scheme on the Cohn-Kanade facial expression database. The model achieves a 96.26 percent overall recognition rate of facial expressions. Table 7 gives some misclassified examples. The pair sadness and anger is difficult even for a human to recognize accurately. The discrimination of happiness and fear failed because these expressions had a similar motion of the mouth.

The experimental results clearly show that our approach outperforms the other dynamic and static methods. Our approach is quite robust with respect to variations of illumination, as seen from the pictures in Fig. 16. It also performed well with some in-plane and out-of-plane rotated

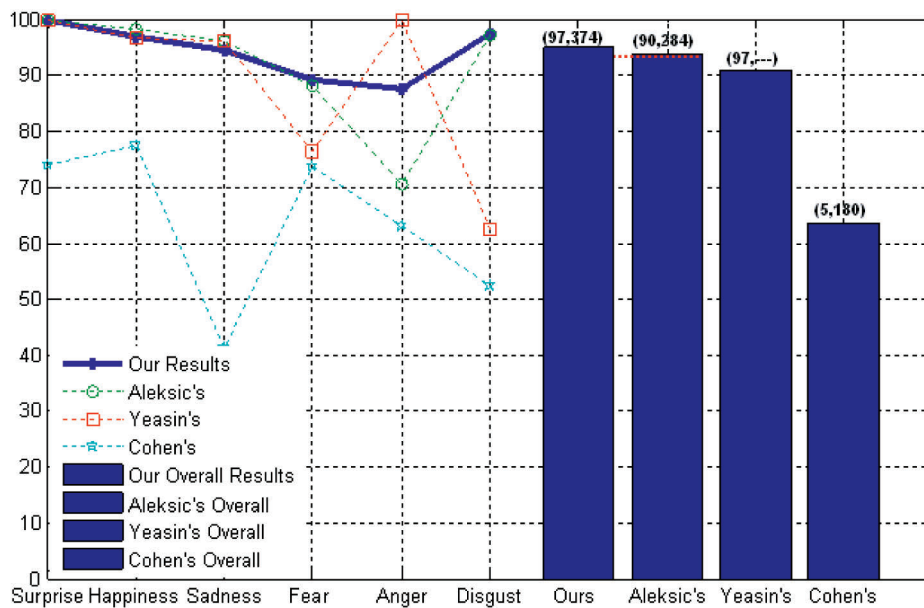


Fig. 18. Comparison with recent dynamic facial expression recognition methods. The left line graphs are recognition results for each expression, whereas the right histograms are the overall rate. The numbers in the brackets express the number of people and sequences used in different approaches. In Yeasin's work, the number of sequences in experiments is not mentioned.

TABLE 4
Results (Percent) of Different Expressions for Two-Fold Cross-Validation

	Surprise	Happiness	Sadness	Fear	Anger	Disgust	Total
$LBP - TOP_{4,4,4,3,3,3}^{u2}$	98.65	89.11	90.41	83.93	93.75	92.11	91.18
$LBP - TOP_{8,8,8,3,3,3}^{u2}$	100.00	97.03	94.52	85.71	84.38	97.37	94.38
$LBP - TOP_{8,8,1,1,1,1}^{u2}$	97.30	90.10	89.04	82.14	84.38	97.37	90.37
$LBP - TOP_{16,16,16,3,3,3}^{u2}$	98.65	91.10	89.04	76.80	75.00	92.10	88.77
$VLBP_{3,2,3}$	95.95	94.06	89.04	83.93	87.50	92.10	91.18
$LBP - TOP_{8,8,8,3,3,3}^{u2} + VLBP_{3,2,3}$	100.00	97.03	94.52	89.29	87.50	97.37	95.19

TABLE 5
Comparison with Different Approaches

	PeopleNum	SequenceNum	ClassNum	Dynamic	Measure	Recognition Rate (%)
[11]	96	320	7(6)	N	10-fold	88.4 (92.1)
[12]	90	313	7	N	10-fold	86.9
[13]	90	313	7	N	leave-one-subject-out	93.8
[38]	97	375	6	N	—	93.8
[14]	97	—	6	Y	five-fold	90.9
[15]	90	284	6	Y	—	93.66
Ours	97	374	6	Y	two-fold	95.19
Ours	97	374	6	Y	10-fold	96.26

sequences. This demonstrates robustness to errors in alignment.

7 CONCLUSION

A novel approach to DT recognition was proposed. A volume LBP method was developed to combine the motion and appearance together. A simpler LBP-TOP operator based on concatenated LBP histograms computed from three orthogonal planes was also presented, making it easy to extract co-occurrence features from a larger number of neighboring points. Experiments on two DT databases with a comparison

to the state-of-the-art results showed that our method is effective for DT recognition. Classification rates of 100 percent and 95.7 percent using VLBP and 100 percent and 97.1 percent using LBP-TOP were obtained for the MIT and DynTex databases, respectively, using more difficult experimental setups than in the earlier studies. Our approach is computationally simple and robust in terms of gray-scale and rotation variations, making it very promising for real application problems. The processing is done locally, catching the transition information, which means that our method could also be used for segmenting DTs. With this, the problems caused by sequences with more than one DT could be avoided.

TABLE 6
Confusion Matrix for $LBP - TOP_{8,8,3,3,3}^{u2} + VLBP_{3,2,3}$ Features

	Surprise 	Happiness 	Sadness 	Fear 	Anger 	Disgust 
Surprise 	98.65		1.35			
Happiness 		96.04		2.97		0.99
Sadness 	1.37		95.89		2.74	
Fear 		3.57	1.79	94.64		
Anger 			3.12		96.88	
Disgust 			2.63		2.63	94.74

TABLE 7
Examples of Misclassified Expressions

Apex image in sequences				
Ground truth	Happiness	Fear	Sadness	Anger
Misclassified results	Fear	Happiness	Anger	Sadness

A block-based method, combining local information from the pixel, region, and volume levels was proposed to recognize specific dynamic events such as facial expressions in sequences. In experiments on the Cohn-Kanade facial expression database, our $LBP - TOP_{8,8,3,3,3}^{u2}$ feature gave an excellent recognition accuracy of 94.38 percent in the two-fold cross validation. Combining this feature with the $VLBP_{3,2,3}$ resulted in further improvement achieving 95.19 percent in twofold and 96.26 percent in 10-fold cross validation, respectively. These results are better than those obtained in earlier studies, even though, in our experiments, we used simpler image preprocessing and a larger number of people (97) and sequences (374) than most of the others have used. Our approach was shown to be robust with respect to errors in face alignment, and it does not require error-prone segmentation of facial features such as lips. Furthermore, no gray-scale normalization is needed prior to applying our operators to the face images.

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